

# CART

## Classification and Regression Trees

*AMS 580 - 2026 edition*

Adapted from the 2025 lecture deck

# Outlines

- ID3
  - i. Key idea
  - ii. Example
- CART
  - i. Key idea
  - ii. Classification tree
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# ID3 (Iterative Dichotomiser 3)

- The original algorithm proposed by Ross Quinlan in 1986 <https://www.chessprogrammi>
- It is a non-linear predictive model.
- In ID3, the leaves represent class labels and branches represent features.
- Both the response and the predictors are categorical variables.



# Example for ID3

| Day | Outlook  | Temperature | Humidity | Wind   | <i>Play Tennis?</i><br>(Y) |
|-----|----------|-------------|----------|--------|----------------------------|
| 1   | Sunny    | Hot         | High     | Light  | No                         |
| 2   | Sunny    | Hot         | High     | Strong | No                         |
| 3   | Overcast | Hot         | High     | Light  | Yes                        |
| 4   | Rain     | Mild        | High     | Light  | Yes                        |
| 5   | Rain     | Cool        | Normal   | Light  | Yes                        |
| 6   | Rain     | Cool        | Normal   | Strong | No                         |
| 7   | Overcast | Cool        | Normal   | Strong | Yes                        |
| 8   | Sunny    | Mild        | High     | Light  | No                         |
| 9   | Sunny    | Cool        | Normal   | Light  | Yes                        |
| 10  | Rain     | Mild        | Normal   | Light  | Yes                        |
| 11  | Sunny    | Mild        | Normal   | Strong | Yes                        |
| 12  | Overcast | Mild        | High     | Strong | Yes                        |
| 13  | Overcast | Hot         | Normal   | Light  | Yes                        |
| 14  | Rain     | Mild        | High     | Strong | No                         |

- Issues when you build up a decision tree:
  1. How to decide the root?
  2. How to specify the test condition?
  3. How to determine the best split?
  4. Determine when to stop the splitting?

# Information: Shannon's entropy

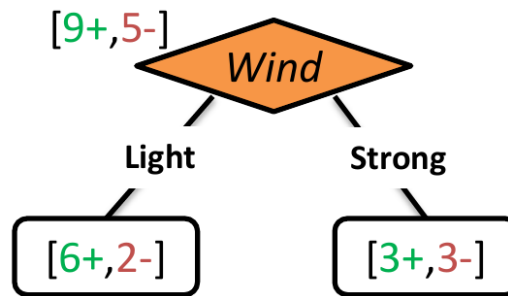
- In information theory, systems are modeled by a transmitter, channel, and receiver. The transmitter produces messages that are sent through the channel. The channel modifies the message in some way. The receiver attempts to infer which message was sent. In this context, **entropy** (more specifically, **Shannon entropy**) is the expected value (average) of the information contained in each message. 'Messages' can be modeled by any flow of information.

- **Shannon entropy:**

$$H(x) = - \sum_{i=1}^n p(x_i) \log p(x_i)$$

# Build up a decision tree

## 1. Selecting the root (node) attribute



Take *Wind* as an example, before we split the node, there are 9+ (Yes) and 5- (No). After splitting, the node with “Light Wind” has 6+ (Yes) and 2- (No), and the one with “Strong Wind” has 3+ (Yes) and 3- (No).

# Build up a decision tree

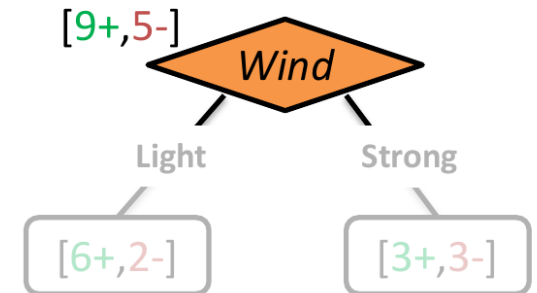
2. Compute the unconditional distribution information:

$$H(p, n) = -\frac{p}{p+n} \log \frac{p}{p+n} - \frac{n}{p+n} \log \frac{n}{p+n}$$

$$- P(\text{Yes}) = \frac{p}{p+n} = \frac{9}{9+5} = \frac{9}{14}$$

$$P(\text{No}) = \frac{n}{p+n} = \frac{5}{9+5} = \frac{5}{14}$$

$$\begin{aligned} - H_{\text{Wind}} &= H(p, n) = H(9, 5) \\ &= -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = \mathbf{0.94 \text{ bits}} \end{aligned}$$

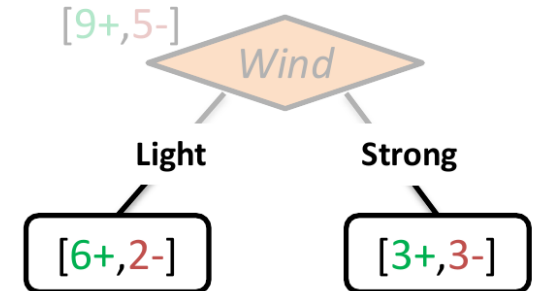


# Build up a decision tree

3. Compute the expected information for subtrees:

$$H(\text{descendant}) = \sum_{i=1}^v \frac{p_i + n_i}{p + n} H(p_i, n_i)$$

- $v = 2$
- $p_1 = 6, n_1 = 2, H(6,2) = 0.81$   
 $p_2 = 3, n_2 = 3, H(3,3) = 1$



$$H(\text{Wind}_{\text{descendant}}) = \frac{6 + 2}{9 + 5} H(6,2) + \frac{3 + 3}{9 + 5} H(3,3) = \mathbf{0.89}$$

# Build up a decision tree

## 4. Finally, compute the information gain:

$$\begin{aligned}
 IG(Outlook) &= H_{Outlook} - H(Outlook_{descendant}) = \mathbf{0.246} \\
 IG(Temperature) &= H_{Temperature} - H(Temperature_{descendant}) = 0.029 \\
 IG(Humidity) &= H_{Humidity} - H(Humidity_{descendant}) = 0.151 \\
 IG(Windy) &= H_{Windy} - H(Windy_{descendant}) = \mathbf{0.05}
 \end{aligned}$$

Since the information gain(IG) for outlook feature is the largest, we take the outlook feature as the root node.

## 5. Do a recursive step here:

```

graph TD
    Sunny --> Node1["{D1, D2, D8, D9, D11}  
[6+, 2-]"]
    Node1 --> Q["?"]
        
```

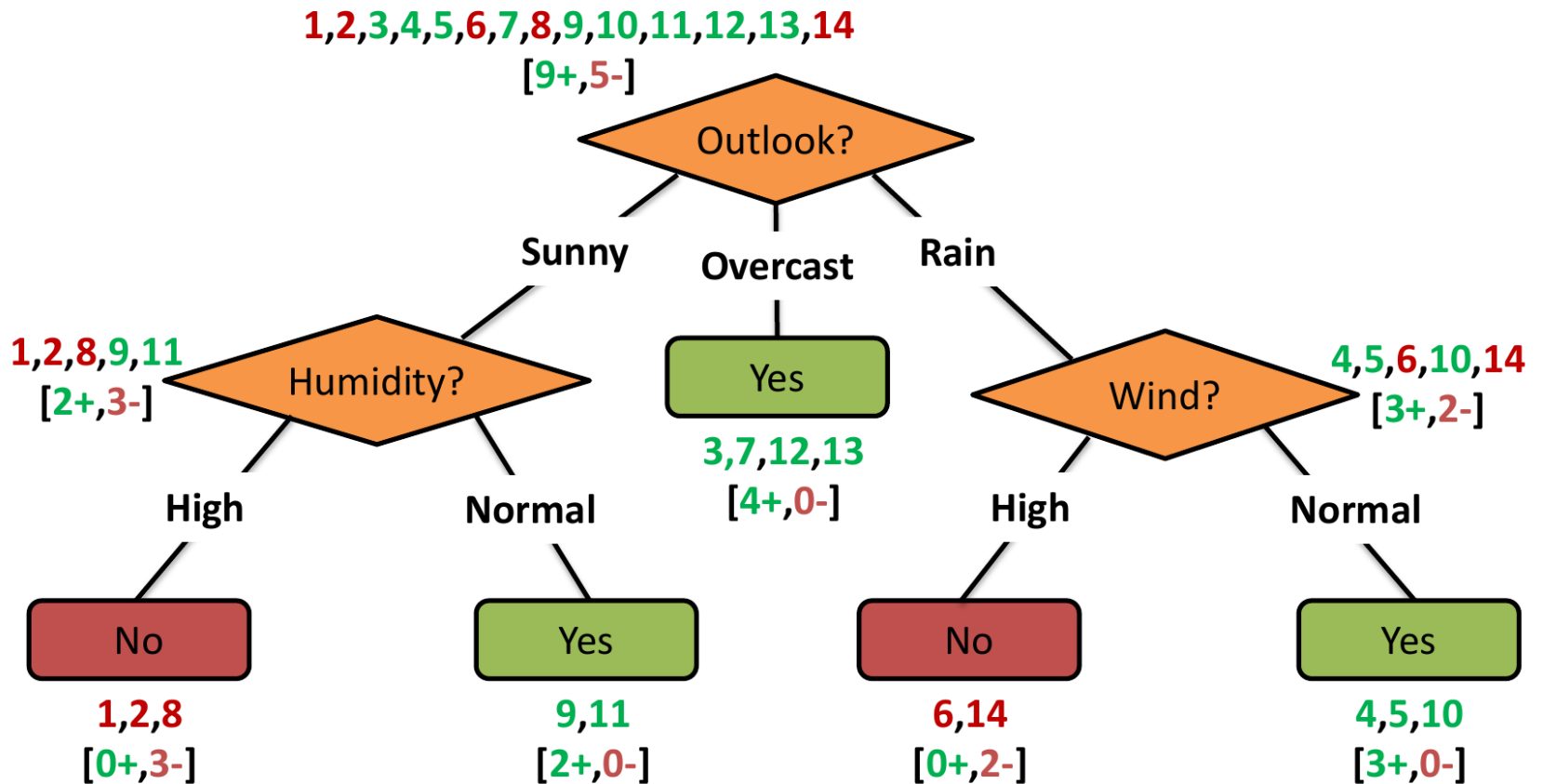
$$S_{sunny} = \{D1, D2, D8, D9, D11\}$$

$$IG(Humidity|S_{sunny}) = H_{Humidity|S_{sunny}} - H(Humidity|S_{sunny\_descendant}) = 0.97$$

$$IG(Temperature|S_{sunny}) = H_{Temperature|S_{sunny}} - H(Temperature|S_{sunny\_descendant}) = 0.57$$

$$IG(Wind|S_{sunny}) = H_{Wind|S_{sunny}} - H(Wind|S_{sunny\_descendant}) = 0.019$$

So we choose Humidity to be the node here.



# Decision tree terminology

- Decision tree builds classification (or regression— as we will see later in CART) models in the form of a tree structure. It breaks down a data set into smaller and smaller subsets while at the same time an associated decision tree is incrementally developed. The final result is a tree with decision nodes and leaf nodes.
- A decision node has two or more branches.
- Leaf node represents a classification or decision.
- The topmost decision node in a tree which corresponds to the best predictor is called the root node.

# CART

## (Classification and Regression Tree)

CART was introduced by

- Leo Breiman (Berkeley),
- Jerome Friedman (Stanford),
- Richard A. Olshen (Stanford), and
- Charles J. Stone (Berkeley), in 1984.



# Philosophy

- “Our philosophy in data analysis is to look at the data from a number of different viewpoints. Tree structured regression offers an interesting alternative for looking at regression type problems. It has sometimes given clues to data structure not apparent from a linear regression analysis. Like any tool, its greatest benefit lies in its intelligent and sensible application.”

--Breiman, Friedman, Olshen, Stone

**Reference:** Breiman, Leo; Friedman, J. H.; Olshen, R. A.; Stone, C. J. (1984). *Classification and Regression Trees*. Monterey, CA: Wadsworth & Brooks/Cole Advanced Books & Software. [ISBN 978-0-412-04841-8](#).

# CART: Classification Tree

- Recursive partition (the same as ID3)
- Consider all possible values of all variables (shown in the example)
- Select the variable/value ( $X = t_1$ ) that produces the greatest “separation” in the target.
- ( $X = t_1$ ) is called a “split”, if  $X < t_1$  then send the data to the “left”, otherwise, send data point to the “right”.
- Only use binary splits
- Splitting criteria for **categorical response (classification tree)**:  
**Gini index**

$$GINI_{node}(Node) = 1 - \sum_{c \in classes} [p(c)]^2$$

# Classification Tree: Gini index

For classification trees with a binary response variable:

- $GINI = 1 - [p_{class\_1}]^2 - [p_{class\_2}]^2$

**Example:** class\_1: 😊 class\_2: ☆



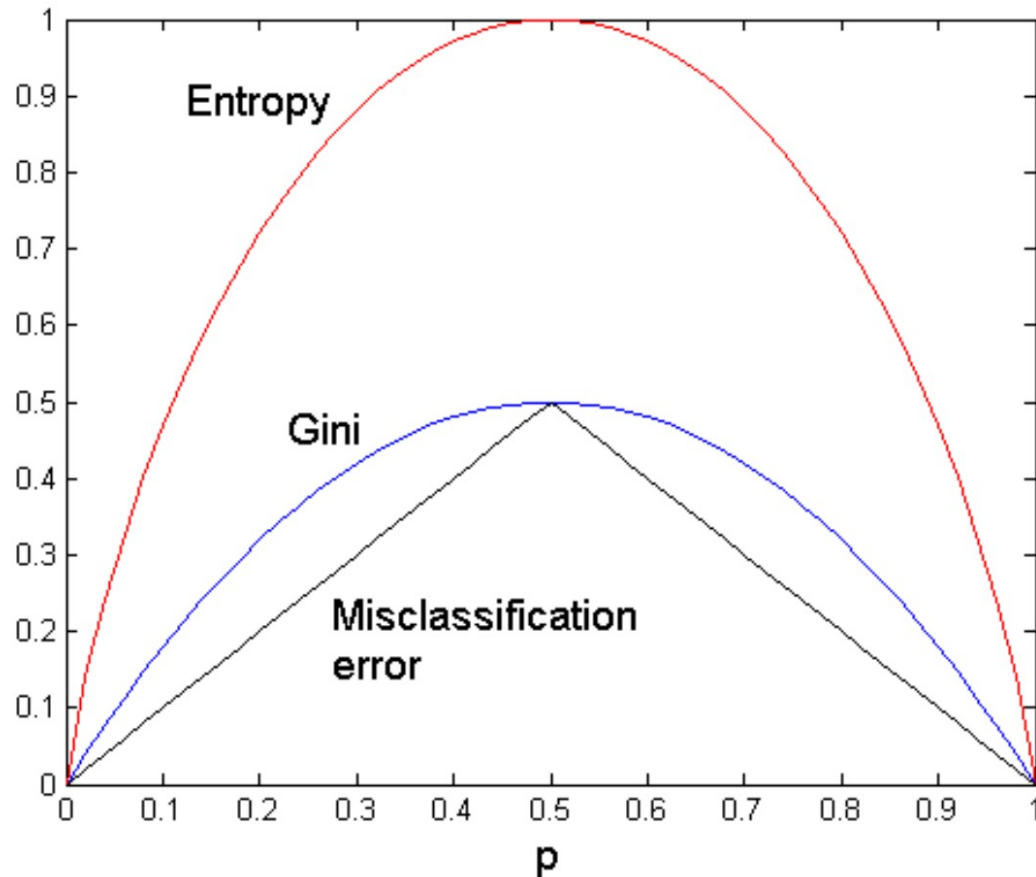
$$p_{class\_1} = \frac{4}{7} \quad \text{and} \quad p_{class\_2} = \frac{3}{7}$$

$$\text{Here, } GINI = 1 - \left[\frac{3}{7}\right]^2 - \left[\frac{4}{7}\right]^2 = 0.48979$$

$$\begin{aligned} \text{Entropy, } H &= -\sum_{i=1}^2 p_{class\_i} \log p_{class\_i} \\ &= -\frac{3}{7} \log \left(\frac{3}{7}\right) - \frac{4}{7} \log \left(\frac{4}{7}\right) = 0.68290 \end{aligned}$$

# Comparison among Splitting Criteria

**For a 2-class problem:**



# CART: Regression Tree: MSE

For regression tree, where the response variable is numerical.  
The splitting criteria for CART is **MSE (mean squared error)**:

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{Y}_i - Y_i)^2$$

Suppose we are generating a tree with two branches. The algorithm first will pick a value, and split the data into two subsets (leaves). For each leaf, it will calculate the MSE.

$$MSE(\text{leaf } k) = \frac{1}{n_k} \sum_{i=1}^{n_k} (\hat{Y}_i - Y_i)^2$$
$$MSE(\text{tree}) = MSE(\text{leaf } 1) + MSE(\text{leaf } 2)$$

In this simple example, the  $\hat{Y}_i$  for each leaf is simple the average of all  $Y_i$  in that leaf. The tree chooses the value with smallest MSE value to split the tree.

# CART

- Suppose you have 3 variables:  
    # vehicles:       $\{1,2,3,\dots,10^+\}$   
    Age category:    $\{1,2,3,\dots,6\}$   
    Liability-only:    $\{0,1\}$
- At each iteration, CART tests all 15 splits.  
     $(\#veh < 2), (\#veh < 3), \dots, (\#veh < 10)$   
     $(age < 2), \dots, (age < 6)$   
     $(lia < 1)$

Select split with the greatest increase in *purity* or with the smallest *MSE*.

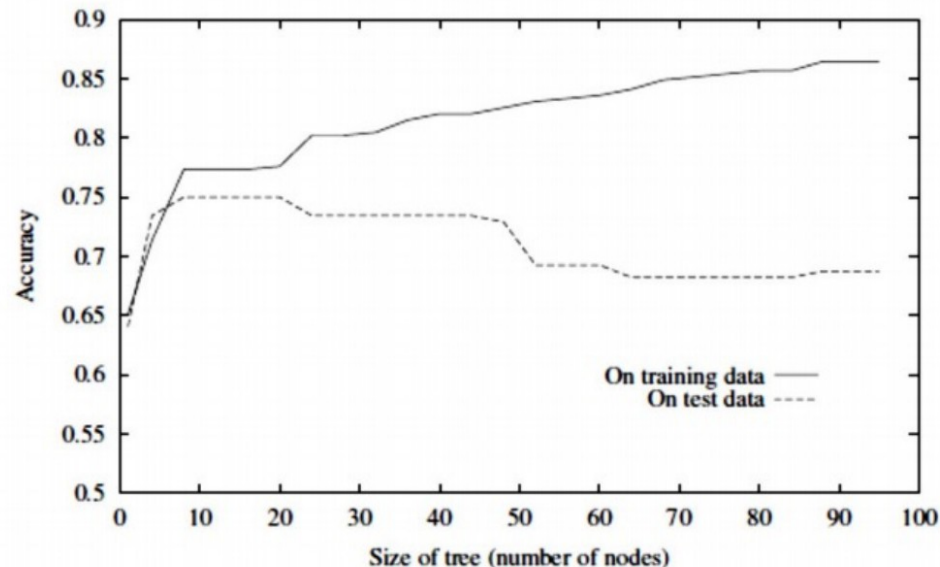
- The algorithm to build a tree is similar to the ID3.

# Comparison between ID3 and CART

- Tests in CART are always binary, but ID3 allows two or more outcomes.
- CART uses the Gini diversity index or the MSE to rank tests, whereas ID3 uses information-based criteria.
- CART can handle both categorical and numerical variables, while ID3 can only handle categorical variables.

# Disadvantages of Trees

- Instability: Even a small change in input data can at times, cause large changes in the tree.
- Complexity: is proportional to  $n$ , which is the size of the feature vector
- Over-fitting:



# Pruning the tree

- Pruning is done by cross validation and finding a tuning parameter  $\alpha$  which makes the cost complexity criterion of the pruned tree less than the cost complexity criterion of the full tree.

$$C_{\alpha}(N) = MSE(tree) + \alpha |N|,$$

where  $|N|$  is number of leaf nodes in the tree.

Note this is the L1-penalty.

- Here we use the notion of “regularization”, namely considering both the model goodness-of-fit and the model complexity (size) together in the fitting criterion:

[https://en.wikipedia.org/wiki/Regularization\\_\(mathematics\)](https://en.wikipedia.org/wiki/Regularization_(mathematics))

- In modern statistics, we have the related branch of “penalized regression”. It can be viewed as a modern variable selection method too. Please study the related sites:

[https://en.wikipedia.org/wiki/Regularized\\_least\\_squares](https://en.wikipedia.org/wiki/Regularized_least_squares)

<http://www.sthda.com/english/articles/37-model-selection-essentials-in-r/153-penalized-regression-essentials-ridge-lasso-elastic-net/>

# Classification tree example

- Wisconsin breast cancer dataset- a very popular dataset for machine learning
- Dr. William Wolberg, Oncology department, University of Wisconsin hospital, collected the data between years 1989-1991 in intervals
- There are 699 instances over 10 attributes to decide if cancer is benign or malignant. 16 missing values.

## **ACKNOWLEDGEMENT:**

The data set has been taken from UCI Machine Learning Repository (where you can find data for your project too):

<http://archive.ics.uci.edu/ml>

# Attributes and outcome

1. Sample code number: id number
2. Clump Thickness: Rating (1-10)
3. Uniformity of Cell Size: Rating (1-10)
4. Uniformity of Cell Shape: Rating (1-10)
5. Marginal Adhesion: Rating (1-10)
6. Single Epithelial Cell Size: Rating (1-10)
7. Bare Nuclei: Rating (1-10)
8. Bland Chromatin: Rating (1-10)
9. Normal Nucleoli: Rating (1-10)
10. Mitoses: Rating (1-10)
11. Class: (2 for benign, 4 for malignant) – (Y)

# SAS PROCEDURE

**The SAS System**

| Obs | code    | thickness | cellsize | cellshape | margadh | epthcellsiz | barnuc | blndchrom | nornuc | mitoses | cancertype |
|-----|---------|-----------|----------|-----------|---------|-------------|--------|-----------|--------|---------|------------|
| 1   | 1000025 | 5         | 1        | 1         | 1       | 2           | 1      | 3         | 1      | 1       | 2          |
| 2   | 1002945 | 5         | 4        | 4         | 5       | 7           | 10     | 3         | 2      | 1       | 2          |
| 3   | 1015425 | 3         | 1        | 1         | 1       | 2           | 2      | 3         | 1      | 1       | 2          |
| 4   | 1016277 | 6         | 8        | 8         | 1       | 3           | 4      | 3         | 7      | 1       | 2          |
| 5   | 1017023 | 4         | 1        | 1         | 3       | 2           | 1      | 3         | 1      | 1       | 2          |
| 6   | 1017122 | 8         | 10       | 10        | 8       | 7           | 10     | 9         | 7      | 1       | 4          |
| 7   | 1018099 | 1         | 1        | 1         | 1       | 2           | 10     | 3         | 1      | 1       | 2          |
| 8   | 1018561 | 2         | 1        | 2         | 1       | 2           | 1      | 3         | 1      | 1       | 2          |
| 9   | 1033078 | 2         | 1        | 1         | 1       | 2           | 1      | 1         | 1      | 5       | 2          |
| 10  | 1033078 | 4         | 2        | 1         | 1       | 2           | 1      | 2         | 1      | 1       | 2          |

# SAS Procedure & RESULTS

```
ods graphics on;  
proc hpsplit data=breastcancer seed=100;  
  class cancertype;  
  model cancertype = thickness cellsize cellshape margadh  
    epthcellsiz barnuc blndchrom nornuc mitoses;  
  grow entropy;  
  prune costcomplexity;  
run;
```

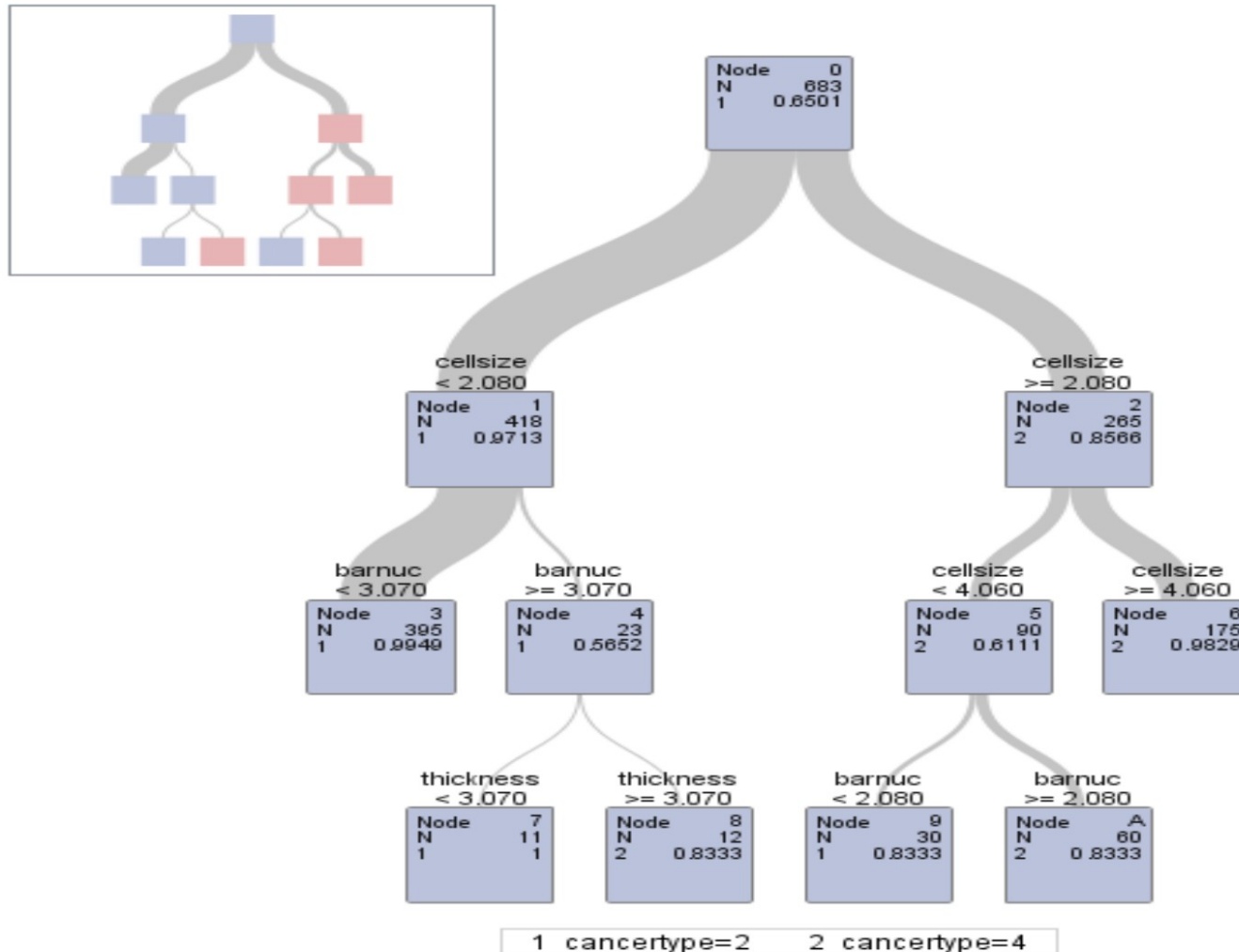
- SAS can grow and prune and give all results together.
- Cross validation is done automatically.

| Model Information               |                 |
|---------------------------------|-----------------|
| Split Criterion Used            | Entropy         |
| Pruning Method                  | Cost-Complexity |
| Subtree Evaluation Criterion    | Cost-Complexity |
| Number of Branches              | 2               |
| Maximum Tree Depth Requested    | 10              |
| Maximum Tree Depth Achieved     | 8               |
| Tree Depth                      | 3               |
| Number of Leaves Before Pruning | 29              |
| Number of Leaves After Pruning  | 6               |
| Model Event Level               | 2               |

|                             |     |
|-----------------------------|-----|
| Number of Observations Read | 699 |
| Number of Observations Used | 683 |

# SAS RESULTS (Breast cancer data)

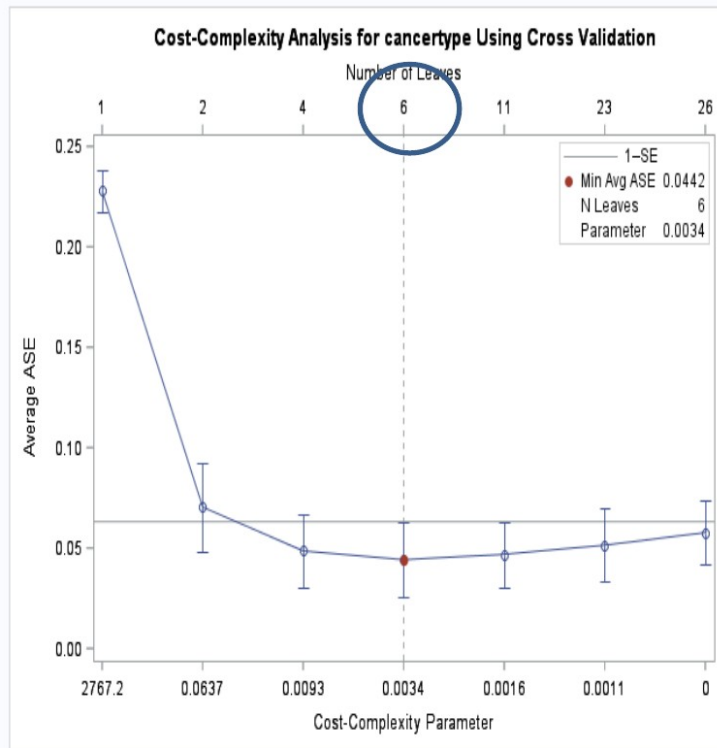
Subtree Starting at Node=0



# SAS Output

The SAS System

The HPSPLOT Procedure



| Model-Based Confusion Matrix |           |     |            |
|------------------------------|-----------|-----|------------|
|                              | Predicted |     | Error Rate |
|                              | 2         | 4   |            |
| Actual 2                     | 429       | 15  | 0.0338     |
| Actual 4                     | 7         | 232 | 0.0293     |

1  
Specificit  
y

1  
Sensitivit  
y

| Model-Based Fit Statistics for Selected Tree |        |           |             |             |         |        |         |        |
|----------------------------------------------|--------|-----------|-------------|-------------|---------|--------|---------|--------|
| N Leaves                                     | ASE    | Mis-class | Sensitivity | Specificity | Entropy | Gini   | RSS     | AUC    |
| 6                                            | 0.0280 | 0.0322    | 0.9662      | 0.9707      | 0.1557  | 0.0559 | 38.2102 | 0.9867 |

# R procedure

- Two common packages for CART in r are:

1) ***tree***

2) ***rpart***

- ***tree*** package is simpler, and has been used here as an example.
- ***tree*** uses the *deviance* (MSE) criteria for splitting. Default splitting criteria for tree package is deviance. Alternative is *Gini index*.
- Pruning has to be done separately using cross validation unlike SAS.
- ***rpart*** is the most popular CART package in r – it has more options and we will go through the following example with ***rpart***:

<http://www.sthda.com/english/articles/35-statistical-machine-learning-essentials/141-cart-model-decision-tree-essentials/>

# R procedure & results

```
> fit<-tree(cancer.type~clmp_thk+un.cell.sz+un.cell.sh+mrg.adh+sngl.cell.sz  
+          +bare.nuclei+blnd.chromatin+normal.nucleoli+mitoses,split="deviance", data=br.  
can)  
>  
> summary(fit)
```

Classification tree:

```
tree(formula = cancer.type ~ clmp_thk + un.cell.sz + un.cell.sh +  
      mrg.adh + sngl.cell.sz + bare.nuclei + blnd.chromatin + normal.nucleoli +  
      mitoses, data = br.can, split = "deviance")
```

Variables actually used in tree construction:

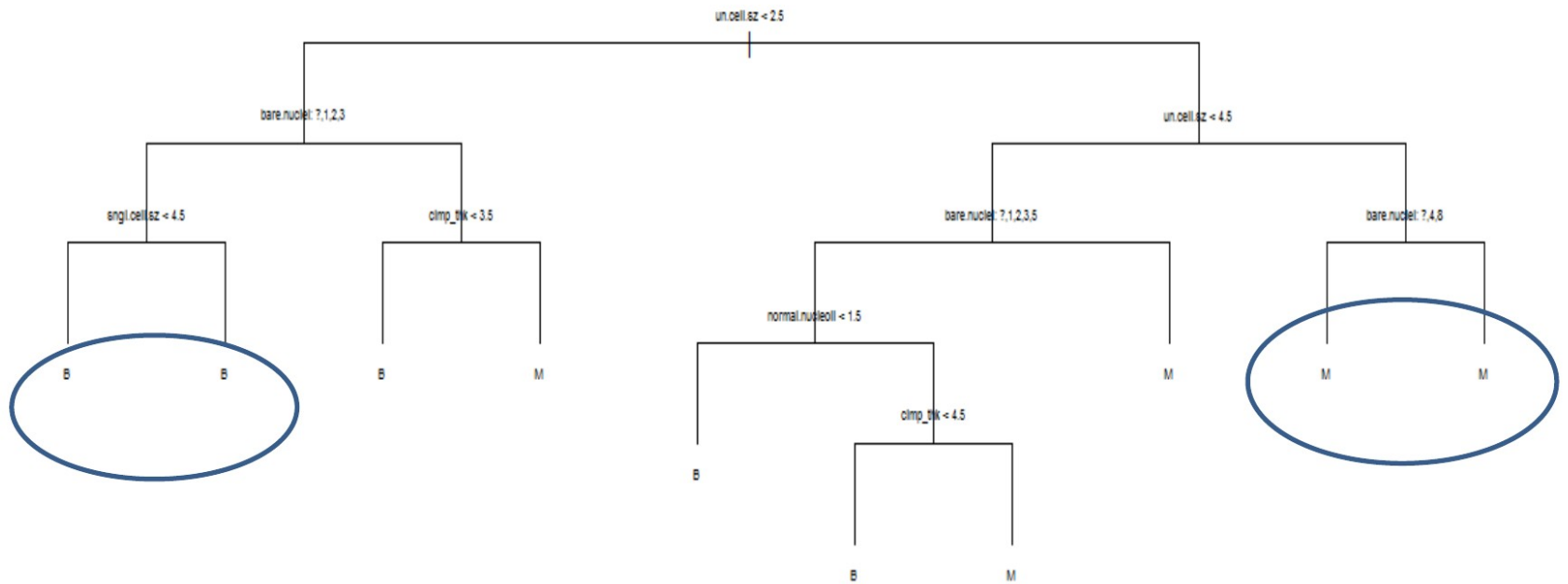
```
[1] "un.cell.sz"      "bare.nuclei"     "sngl.cell.sz"    "clmp_thk"  
[5] "normal.nucleoli"
```

Number of terminal nodes: 10

Residual mean deviance: 0.1444 = 99.47 / 689

Misclassification error rate: 0.02718 = 19 / 699

Observe!!



# Cross validation to prune the tree

```
> set.seed(100)
> cv.tree(fit)
$size
[1] 10  9  8  7  6  5  4  3  2  1

$dev
[1] 398.4250 403.6903 350.0134 354.3141 353.6841 356.6726 365.1357 365.0388 348.3362
[10] 902.4477

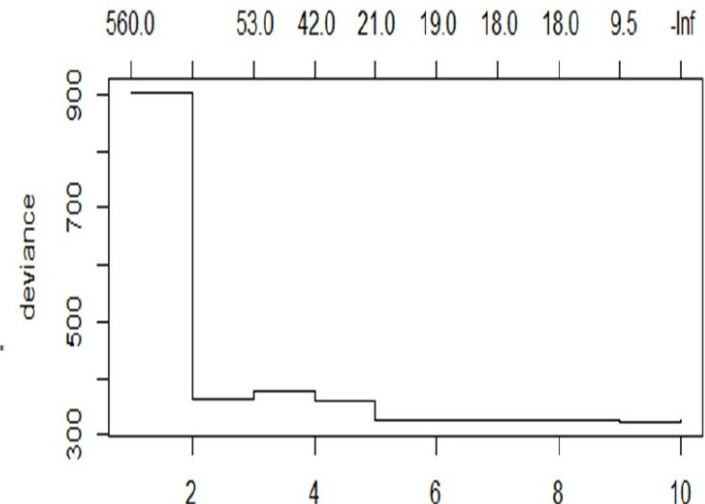
$k
[1] -Inf  9.505041 17.604785 18.035461 18.548001 20.678880 41.627949
[8] 52.763064 61.252252 561.038804

$method
[1] "deviance"

attr(,"class")
[1] "prune"      "tree.sequence"
> plot(cv.tree(fit))
> prune_fit = prune.misclass(fit, best = 7)
> plot(prune_fit,type="uniform")
> text(prune_fit,pretty=0,cex=0.5)
> summary(prune_fit)
```

```
Classification tree:
snip.tree(tree = fit, nodes = c(4L, 7L))
Variables actually used in tree construction:
[1] "un.cell.sz"      "bare.nuclei"    "clmp_thk"
Number of terminal nodes: 8
Residual mean deviance:  0.1963 = 135.6 / 691
Misclassification error rate: 0.02718 = 19 / 699
```

"normal.nucleoli"



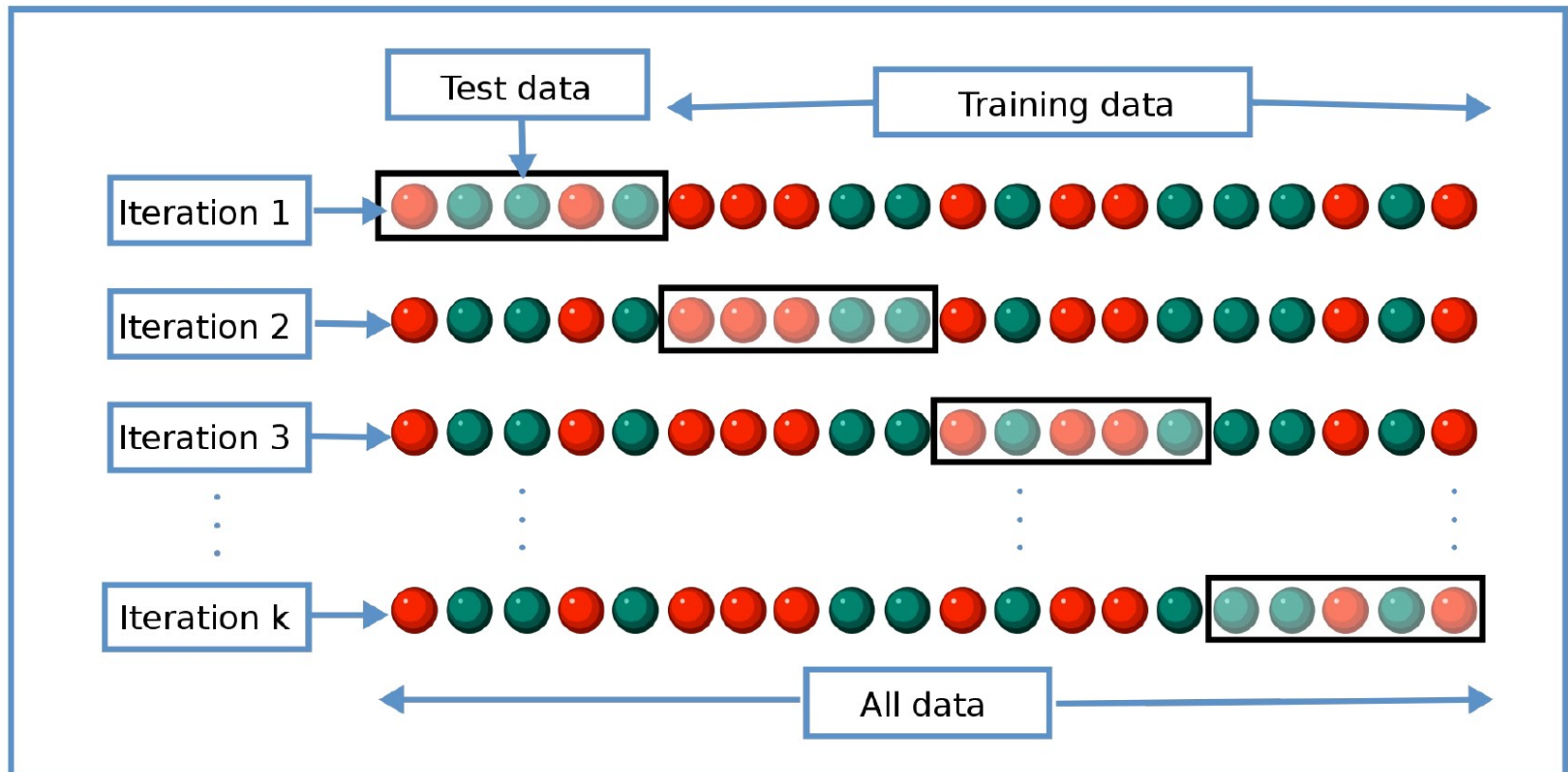
size

# Cross-Validation

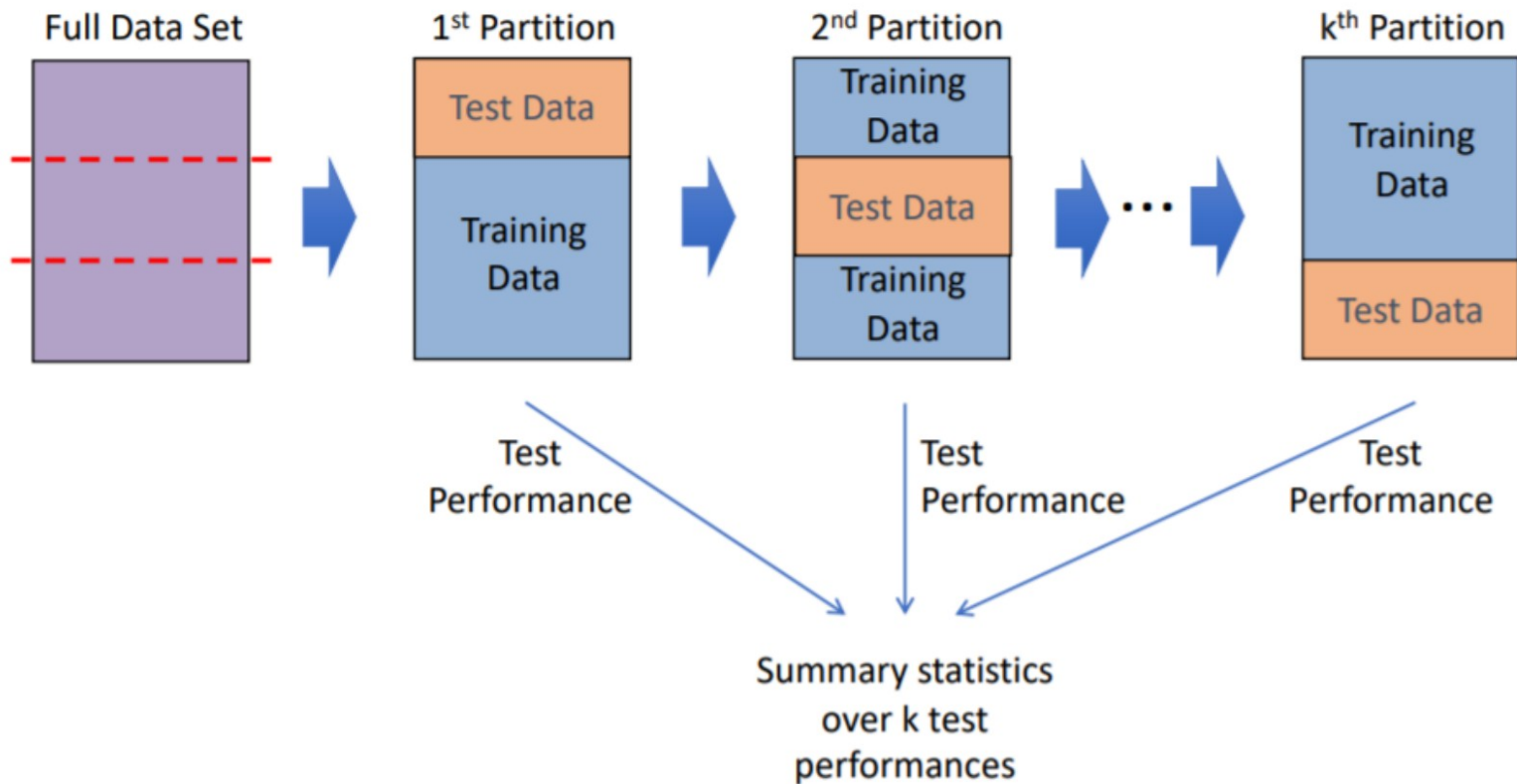
- ***K-fold cross-validation*** avoids overlapping test sets
  - First step: split (the training) data into  $k$  subsets of equal size
  - Second step: use each subset in turn for testing, the remainder for training
  - This means the learning algorithm is applied to  $k$  different training sets
- Often the subsets are stratified before the cross-validation is performed to yield stratified  $k$ -fold cross-validation
- The error estimates are averaged to yield an overall error estimate; also, standard deviation is often computed

# K-Fold Cross Validation

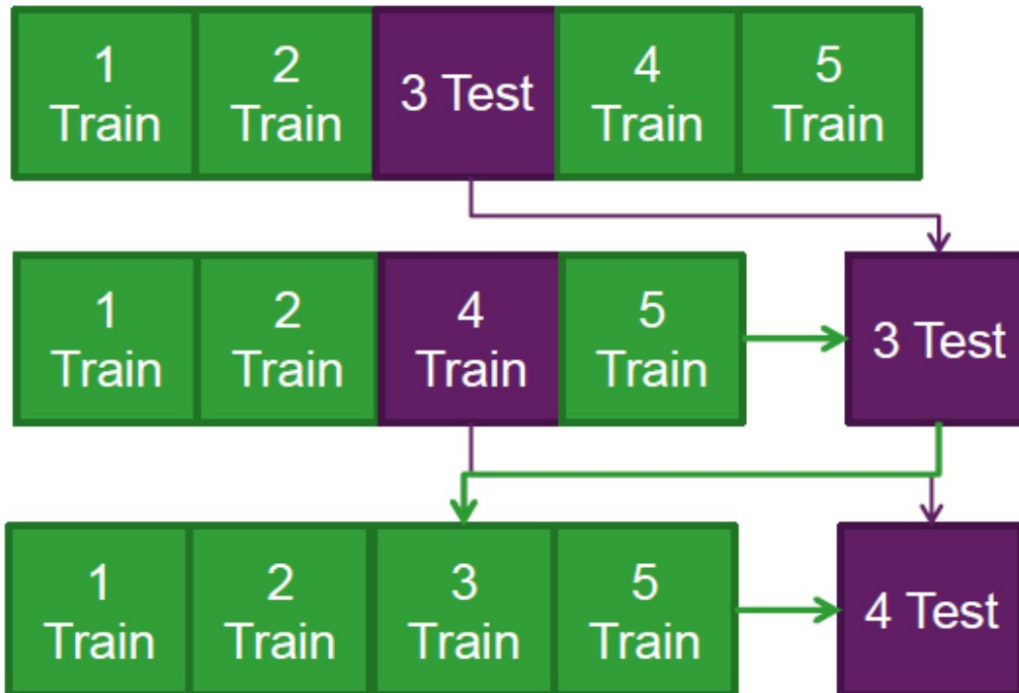
[https://en.wikipedia.org/wiki/Cross-validation\\_\(statistics\)#/media/File:K-fold\\_cross\\_validation\\_EN.svg](https://en.wikipedia.org/wiki/Cross-validation_(statistics)#/media/File:K-fold_cross_validation_EN.svg)



# K-Fold Cross Validation



# 5-Fold Cross Validation



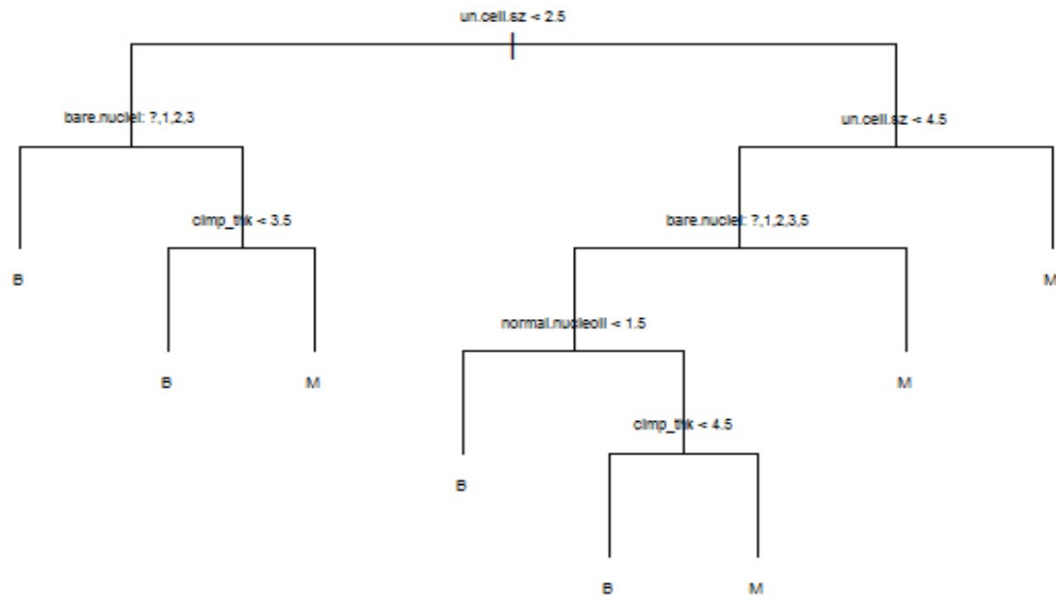
1. Split the training data into 5 samples
2. Fit a model to the training samples and use the test sample to calculate a CV metric.
3. Repeat the process for the next sample, until all samples have been used to either train or test the model

# Leave-One-Out Cross-Validation

- **Leave-one-out** (also referred to as “**Jackknife**”) cross-validation is a special (extreme) form of  $k$ -fold cross-validation:
  - Set number of folds to number of training instances:  
 $k = \text{number of observations } (n) \text{ in the training data}$
  - That is, for  $n$  training instances, build classifier  $n$  times
- Makes best use of the data
- Involves no random subsampling
- Computationally expensive



# The pruned tree is simpler and hopefully more robust



# Regression tree (example in R)

```
> head(housing)
  MedianHouseValue MedianIncome MedianHouseAge TotalRooms TotalBedrooms
1         452600         8.3252           41         880         129
2         358500         8.3014           21        7099        1106
3         352100         7.2574           52        1467         190
4         341300         5.6431           52        1274         235
5         342200         3.8462           52        1627         280
6         269700         4.0368           52         919         213

  Population Households Latitude Longitude
1         322         126      37.88    -122.23
2        2401        1138      37.86    -122.22
3         496         177      37.85    -122.24
4         558         219      37.85    -122.25
5         565         259      37.85    -122.25
6         413         193      37.85    -122.25
```

- ACKNOWLEDGEMENT:

You can find the CA housing data as well as the r program from the following website:

[https://rstudio-pubs-static.s3.amazonaws.com/304866\\_e75f606e64824e4789cde7bd30ea52ae.html](https://rstudio-pubs-static.s3.amazonaws.com/304866_e75f606e64824e4789cde7bd30ea52ae.html)

This data set is widely used also, for example in homework assignments given by Prof. Cosma Shalizi, CMU, Statistics department. <http://www.stat.cmu.edu/~cshalizi/350/>

# Advantages of CART

- Very easy to interpret and implement
- Can work with many covariates
- Can handle and explain complex interactions (non-linear relations too) between covariates.
- Do not have to clean the data too much, can handle missing values in SAS/R- surrogate variables
- Decides which covariates are important automatically
- Non parametric
- Computationally fast
- Monotonic transformations of outcome will not affect CART

# Disadvantages of CART

- CART may have unstable decision trees
- A slight perturbation can affect the whole decision tree
- CART splits only on basis of one variable. So the tree can be complex in the end.
- An ensemble of decision trees will almost always outperform a single CART
- Should be used in cases where overfitting is not a problem